Forecasting Indian crude oil Price Using Hybrid ARIMA-GARCH Model.

Overview

- A commonly used forecasting method is the **Box–Jenkins method**, which uses **ARIMA** models.
- It can provide accurate forecasts over short periods, but it struggles to handle the volatility and nonlinearity of oil prices. Thus, this paper proposes a **hybrid ARIMA–GARCH model** to forecast oil prices.
- The data used for forecasting are monthly **Indian crude oil prices from April 2000 to October 2023**. The results show that the **ARIMA(4,2,0)–GARCH(0,3)** hybrid model is the most suitable for forecasting oil prices. The hybrid model provides better forecasts and is expected to be valuable to all stakeholders in the industry

Literature Review

> Factors afecting oil price

- Miao, Ramchander, Wang, and Yang's (2017) paper titled **"Influential Factors in Crude Oil Price Forecasting"** studies the key factors that play a significant role in predicting crude oil prices. It is a valuable contribution to the field of energy as it helps to identify the drivers and determinants that impact crude oil price forecasts.
- The paper by Sultan 2020 titled **"Empirical Investigation of Relationship between Oil Price and Inflation: The Case of India"** discusses the relationship between oil prices and inflation. It states that fluctuation in oil prices impacts inflation rates and provides insights into economic significance

> Forecasting oil price

- Shambulingappa's (2020) titled **"Crude Oil Price Forecasting using Machine Learning"** discusses various machine learning techniques for predicting crude oil

prices. This enables the integration of contemporary methods in forecasting commodities in the energy sector.

- Bhattacharya and Ahmed (2018) titled **"Forecasting Crude Oil Price Volatility in India using a Hybrid ANN-GARCH Model"** discusses an innovative approach to predicting crude oil prices using a hybrid model that combines an artificial neural network (ANN) and generalised autoregressive conditional heteroskedasticity (GARCH). This research has contributed to the Indian crude oil market by providing valuable insights.

Research Method



> Box-Jenkins Method

- The Box–Jenkins Methodology (1976) is an approach for developing and designing ARIMA models for forecasting.
- The Box–Jenkins methodology has four steps: model identification, parameter estimation, diagnostic testing, and forecast evaluation.
- **ARIMA(p,q,d) :** $\phi_p(L)(1-L)^d(y_t-\mu)=\theta_q(L)e_t$

where,

$\phi_p(L) = 1 - \sum_{i=1}^{q} \phi_i L^i$, $\theta_q(L) = 1 - \sum_{i=1}^{q} \theta_i L^i$ are polynomials in terms of L of degrees p an q.

y_t is the time series , e_t is the error term at time t , μ is the mean of the model d is the order of difference ∅*p* and *ϑj* are the parameters of autoregressive and moving average terms L is a difference operator defined as Δ*yt* = *yt* - *yt*−1 = (1-L) *yt*

Grid Search Method

- The Grid Search method is a comprehensive ML approach that rigorously explores a range of potential ARIMA model configurations that vary AR (p), differencing(d), and MA(q) parameters within a specified range. The method aims at minimising criteria such as the Aikaike Information Criterion (AIC) or Bayesian Information Criterion (BIC).

Hetroskedastic Modelling

- The ARIMA model fails to consider the volatility of the variables. Generalized autoregressive conditional heteroskedasticity (GARCH) models are used to model financial time series with varying volatility Bollerslev's(1986) proposed GARCH(r,s) model[13], as given below:

  $y_t = \mu_t + z_t$

  Where, μ_t is conditional mean of y_t Z is the shock at time t $z_t = \sigma_t e_t$ Where e_t → *iid N*(0,1) and $\sigma_t{}^2 = \alpha_0 + \sum_{i=1}^{r} \alpha_i z_{t-1}^2 + \sum_{i=1}^{s} \beta_i \sigma_{t-1}^2$ Where, σ_t^2 is the conditional variance of *yt*, α_0 is a constant term, r is the order of the ARCH terms s is the order of the GARCH terms *αi* and *βi* are the coefficients of the ARCH and GARCH parameters respectively

Hybrid ARIMA-GARCH Model

- The ARIMA model captures only the linear and stationary aspects of the data, whereas the residuals and the remaining nonlinear patterns are isolated. The hybrid ARIMA– GARCH model involves a two-stage process: initially, the ARIMA model is built, and then, in a subsequent stage, the GARCH model is built to analyse the residuals

- GARCH - paper

Statistics

Crude Oil Price	
Mean	65.04
Standard deviation	28.68
Median	62.53
Minimum	18.24
Maximum	132.47
Skewness	0.25
Kurtosis	-0.97
Observations	283

- The results show that the mean oil price (65.04) is quite higher than the standard deviation (28.68) indicating the price volatility. The skewness of 0.25 says that the dataset is slightly positively skewed, implying longer tails, whereas the kurtosis of -0.97 indicates that the distribution has thinner tails than a normal distribution with fewer and fewer extreme outliers

- **Dickey– Fuller (ADF) test** : confirm that crude oil prices are non-stationary in their levels, and the probability value is greater than the significance level of 0.05 (5%). Thus, a differentiation transformation is performed to obtain the stationary time series data.

Crude Oil Price	
Augmented Dickey-Fuller test	
Dickey-Fuller	-2.903
p-value	0.4959
Alternative Hypo-thesis	Stationary

Emperical Results

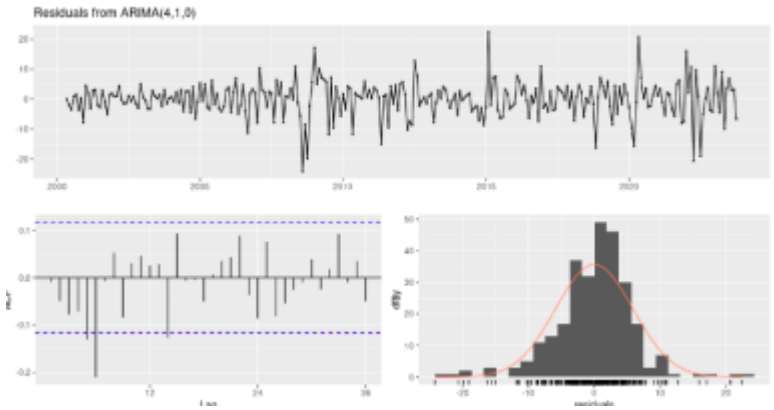
- In consideration of the comprehensive scope, values of p and q are taken up to 20

- The table below shows only the top 5 configurations that exhibited the most favorable model fits.

ARIMA Configuration	AIC
ARIMA (0,1,0)	1855.436
ARIMA (1,1,0)	1831.921
ARIMA (2,1,0)	1824.858
ARIMA (3,1,0)	1810.944
ARIMA (5,1,0)	1805.392
ARIMA (4,1,0)	1803.977

- ARIMA (4,1,0) model is the most suitable for crude oil prices. This configuration implies that a first-order difference was necessary to achieve stationarity, and the model includes AR terms (p=4) without moving average terms (q=0). The AIC for the ARIMA (4,1,0) is 1803.977 and the RMSE is 5.87.

- ARIMA (4,1,0) model. The autocorrelation for residuals indicates the volatility in the dataset and the need for the GARCH model



- From the ACF and PACF plots of the Squared Residuals and Grid method, the most suitable Hybrid model is ARIMA(4,1,0)-GARCH(0,3). The following Table shows the coefficients of the Hybrid Model.

ARIMA (4,1,0)	
	Co-efficient
AR 1	-0.4320
AR2	-0.3069
AR3	-0.3042
AR4	-0.1771
GARCH (0,3)	
	Co-efficient
Mu	33.73
Omega	24.54
Beta 1	5.4e-07
Beta 2	0.00392
Beta 3	0.995

Forecasting

- After obtaining results from the ARIMA (mean modelling) and GARCH (variance modelling), crude oil price is predicted for the next 1 month. From the above modelling results, a model that is good enough to represent Indian crude oil price data is the ARIMA (4,1,0)-GARCG (0,3) model. The forecast for November 2023 is 87.26, which is very close to the actual value of 87.09. This suggests that the model has less than 1% error.